

Additive Manufacturing of High-Entropy Alloys for Marine Applications: Processing, Corrosion Performance, and Pathways to Qualification

Anirban Chakraborty, Pankaj Sharma*, and Chinmoy Sarkar
Indian Maritime University

Abstract—High-entropy alloys (HEAs) combine high strength, structural stability, and—in carefully chosen compositions—passive-film corrosion resistance that can exceed conventional marine alloys in chloride media. Additive manufacturing (AM) is a natural processing partner for these alloys: rapid solidification suppresses detrimental elemental segregation, refines the microstructure, and can improve pitting resistance relative to cast material, while powder- and wire-based routes enable complex wetted components, large structures, and in-service repair that are uneconomical by casting or forging. This review synthesizes the AM processing of HEAs (LPBF, SEBM, DED/LMD, WAAM, and binder jetting), the as-built and heat-treated microstructures that result, and their electrochemical and corrosion performance in saline and seawater environments, before mapping these capabilities onto concrete marine applications: seawater piping, pumps and valves, heat exchangers, propulsion components, and cladding-based repair of shafts and propellers. We show that the main scientific gaps are environment-specific: long-term natural seawater exposure, biofilm-coupled corrosion, corrosion fatigue and stress corrosion cracking data, and galvanic compatibility of HEA deposits on marine steels are all poorly quantified. The main engineering gaps are feedstock cost, defect control, and the absence of classification-society qualification pathways for HEA compositions. We close with a staged roadmap linking alloy design, process control, and standards-based qualification for marine deployment.

Index Terms—High-entropy alloys, additive manufacturing, laser powder bed fusion, wire arc additive manufacturing, directed energy deposition, seawater corrosion, pitting, passivation, marine engineering, qualification.

I. INTRODUCTION AND SCOPE

Corrosion in marine and offshore systems is a first-order economic and safety problem. The global cost of corrosion has been estimated at roughly USD 2.5 trillion per year, on the order of 3–4% of world gross domestic product, with marine infrastructure, shipping, and offshore energy accounting for a substantial share [1]. Seawater is among the most aggressive natural electrolytes: a near-neutral solution containing approximately 0.5–0.6 mol/L of chloride, dissolved oxygen, biological activity, and wide swings in temperature, flow velocity, and aeration between the atmospheric, splash, tidal, and fully immersed zones [2].

Conventional marine alloys each carry well-known limitations. Austenitic 316L stainless steel relies on a chromium-rich passive film but remains susceptible to pitting, crevice corrosion, and stress corrosion cracking (SCC) in warm, stagnant, or

creviced chloride service [3]. Nickel–aluminum bronzes resist biofouling reasonably well but suffer dealuminification and cavitation damage; copper–nickel alloys trade strength for antifouling behavior; duplex stainless steels and titanium alloys perform well but are expensive and difficult to fabricate into complex or large geometries [2]. A material family that combines strength, toughness, and robust passivity in chloride media—and that can be formed into complex, repairable components—would therefore address a genuine gap.

High-entropy alloys (HEAs), defined by five or more principal elements in near-equi-molar or deliberately non-equi-molar ratios [4], [5], are credible candidates. Their high configurational entropy, severe lattice distortion, sluggish diffusion, and the so-called “cocktail” effect together stabilize simple solid-solution phases and, in Cr-bearing compositions, support protective passive films [6]–[8]. Reviews of HEA corrosion behavior identify compositions—particularly FCC CoCrFeMnNi-type and carefully balanced Al-containing eutectic alloys—whose pitting resistance in chloride solutions rivals or exceeds 316L [9], [10].

Additive manufacturing (AM) completes the picture for three reasons. *First*, AM’s rapid solidification ($\sim 10^3$ – 10^6 K/s in powder-bed processes) suppresses the coarse segregation typical of cast HEAs, and recent evidence shows that this homogeneity can translate directly into better chloride corrosion performance [11], [12]. *Second*, AM enables geometries that casting cannot deliver economically: conformally cooled heat exchangers, topologically optimized brackets, and integrated manifolds for seawater systems [13]. *Third*, directed energy deposition (DED) and wire arc additive manufacturing (WAAM) make HEAs practical as repair and cladding materials for large marine components, avoiding the need to build entire structures from expensive alloys [14], [15].

Existing reviews have treated AM of HEAs comprehensively from the processing and mechanical-property viewpoints [12], [13], [16], and marine-corrosion texts mention HEAs as an emerging material class [2], but no synthesis to date is organized around *marine deployment*: which compositions survive real seawater service, which AM routes suit which marine components, and what qualification evidence is missing. This review targets that gap. Its scope is: (i) HEA fundamentals relevant to marine service (Section II); (ii) AM processes and feedstock for HEAs (Section III); (iii) microstructure and mechanical performance of AM HEAs (Section IV); (iv) corrosion and electrochemical behavior in saline and seawater

*Corresponding author. Email: pankajs@imu.ac.in

environments (Section V); (v) candidate marine applications and qualification pathways (Section VI); and (vi) challenges and a deployment roadmap (Sections VII–VIII). Table and figure sets are collected in Section X.

A. Review methodology and literature coverage

To formalize coverage and limit selection bias, the literature search followed a PRISMA-informed protocol [17], executed on 8 October 2026 (Fig. 2). *Sources*: 26 structured queries run through a web-search interface surfacing indexed sources (ScienceDirect, SpringerLink, MDPI, Frontiers, Nature portfolio, Wiley, IOP, ASTM, OSTI, NRC, ResearchGate/Semantic Scholar records), plus reference-list snowballing from six AM-HEA and marine-corrosion reviews. *Keywords*: built from three facets — alloy family (“high-entropy alloy”, “CoCrFeMnNi”, “eutectic high-entropy alloy”, “refractory HEA”, “medium-entropy alloy”), process (“laser powder bed fusion”, “SLM”, “directed energy deposition”, “wire arc additive manufacturing”, “binder jetting”, “laser cladding”), and marine/corrosion service (“seawater”, “chloride”, “pitting potential”, “crevice”, “stress corrosion cracking”, “microbiologically influenced corrosion”, “cathodic protection”, “marine”) — combined with benchmark terms (“316L”, “NiAl bronze”). No date restriction was imposed; records span 2004–2026 with emphasis on 2019–2026. *Inclusion criteria*: peer-reviewed primary studies or verifiable technical reports on AM HEAs (or AM stainless analogs where defect–corrosion links were needed) reporting microstructure, mechanical, electrochemical, corrosion, MIC, or standards content relevant to marine service. *Exclusion criteria*: non-English records; records without verifiable bibliographic identity; duplicate studies superseded by stronger evidence; off-topic HEA applications. Counts as recorded: ≈ 105 records surfaced by queries, ≈ 35 by snowballing, ≈ 120 unique records screened, ≈ 59 excluded, 61 included (see `references.bib`). *Declared limitation*: subscription databases (Scopus/Web of Science) were not accessible from the review environment, so coverage is structured and transparent but not exhaustive; gray-literature records were included only when bibliographically verifiable, and unverifiable details are flagged as such in the Data Appendix.

Sensitivity analysis (sentinel checks): to estimate the probability that major studies were missed despite the database limitation, three sentinel topics most likely to harbor unseen literature were re-queried with 2024–2026 filters and the returns compared against the reference list. (i) *MIC of HEAs*: sentinel returns comprised ~ 5 primary studies plus one review; this review now cites five of five identified primaries plus the review [18]–[22]—coverage assessed as high. (ii) *Natural-seawater field exposure of AM HEAs*: sentinel returns contained no AM-HEA field studies (only conventional-alloy biofilm chronologies such as the 16-week CuNi/Ti microfouling series [23] and AM-aluminum analogues)—the Section IX conclusion that this is a genuine evidence gap is therefore robust to search limitations. (iii) *2025–2026 LPBF-HEA chloride corrosion*: sentinel returns were limited to the EHEA-homogenization and Mo-alloying records already cited [11], [24]. Assessment: low probability that a missed study would overturn a stated conclusion; the

residual risk concentrates in paywalled conference proceedings, which could not be screened.

II. HEA FUNDAMENTALS RELEVANT TO MARINE SERVICE

A. Core effects, critically stated

The original HEA concept invoked four “core effects”: high configurational entropy stabilizing solid solutions, severe lattice distortion impeding dislocation motion and diffusion, sluggish diffusion retarding phase decomposition, and a cocktail effect in which properties emerge non-linearly from the mixture [4], [8]. A critical reassessment by Miracle and Senkov [6] retains configurational entropy as a genuine thermodynamic contribution but cautions that it alone does not guarantee single-phase microstructures, that lattice distortion is composition-dependent, and that “sluggish diffusion” is neither universal nor necessarily large. For the marine engineer the practical restatement is simple: HEA compositions must be evaluated case by case, and phase stability in the presence of defects, segregation, and service temperature is what matters [7].

B. Phase families and their marine relevance

FCC alloys (prototypically CoCrFeMnNi, the “Cantor” alloy [5]) are ductile at all temperatures, with strength rising as temperature falls [25] and fracture toughness that remains exceptional even at cryogenic temperatures [26]. This combination is attractive for seawater piping, cryogenic cargo systems, and any component where corrosion resistance must coexist with toughness. Their weakness is moderate yield strength, which must be supplied by grain refinement, cold work, or precipitation engineering.

BCC and refractory HEAs (e.g., Nb-, Mo-, Ta-, W-rich systems) offer high strength and thermal stability but typically poorer ambient ductility and, depending on composition, weaker passivity; their AM processing is reviewed in [16]. For marine use they are best viewed as candidates for hot or wear-dominated niches rather than bulk seawater-wetted structures.

Eutectic HEAs (EHEAs), led by AlCoCrFeNi_{2.1} with its lamellar FCC/B2 duplex structure [27], balance strength and ductility and—importantly for AM—solidify as a coupled eutectic, which reduces hot cracking susceptibility. Recent work shows that the corrosion behavior of this family is exquisitely sensitive to how uniformly the elements partition between the two phases [11], making AM’s rapid solidification a genuine processing lever.

C. Passivity and its limits in chloride media

HEA corrosion resistance is not automatic. The most influential corrosion review of HEAs, by Qiu et al. [9], established that Cr-rich passive films are the basis of good performance, while Al- and Ni-rich BCC/B2 phases in multiphase alloys are preferential pitting sites; later work on Al_xCoCrFeNiTi_y quantified how microstructural evolution shifts electrochemical response [10]. Two directions are directly relevant to seawater service. First, *computational alloy design for passivity*: Lu et al. [28] demonstrated a thermodynamics-plus-electrochemistry

design loop that produced a corrosion-resistant HEA for harsh environments, and Quiambao et al. [29] characterized its passivation in detail, showing that selective Cr and Fe enrichment of the passive film underlies its resistance. Second, *Mo and W additions* strengthen passive films in chloride solutions in much the same way Mo does in stainless steels, a strategy recently exploited in EHEAs (Section V).

Two marine-specific caveats follow. Passive-film stability is *environment-specific*: an alloy that passivates in dilute or neutral chloride can behave very differently in acidic, sulfide-bearing, or creviced conditions typical of stagnant seawater or under-biofilm niches. And the “cocktail effect” works both ways: beneficial synergies are documented, but adverse ones—e.g., micro-galvanic coupling between constituent phases—are equally possible [2], [9].

III. ADDITIVE MANUFACTURING ROUTES FOR HEAS

AM of HEAs has been reviewed comprehensively by Cagirici et al. [13] and Han et al. [12]; here we summarize the process-property consequences that matter for marine components, then discuss feedstock, which is the binding constraint for marine-scale deployment. Table IX (Section X) compares the routes.

A. Powder bed fusion: LPBF and SEBM

Laser powder bed fusion (LPBF) melts thin powder layers with a focused laser, producing the highest cooling rates ($\sim 10^5$ – 10^6 K/s) and the finest as-built microstructures of any practical HEA route: cellular and dendritic substructures with sub-micron dislocation cells and strong crystallographic texture along the build direction [12], [13]. The same rapid solidification that refines the structure is what suppresses elemental segregation in multiphase HEAs; this is the microstructural origin of the improved chloride corrosion resistance reported for LPBF CoCrFeMnNi and SLM AlCoCrFeNi_{2.1} relative to cast material [11], [30]. The penalties are limited build volume, slow build speed, and susceptibility to lack-of-fusion and keyhole porosity when parameters are off-window; high-Al compositions additionally face solidification cracking [13].

Selective electron beam melting (SEBM/EBAM-class processes) operates under vacuum with a heated powder bed, reducing thermal gradients and residual stress. This makes SEBM the preferred powder-bed route for crack-prone and refractory systems [16], and its preheat capability is valuable for BCC HEAs with ductile-to-brittle transitions. Surface finish and resolution are coarser than LPBF, so wetted surfaces intended for corrosion service require machining or polishing.

B. Directed energy deposition and laser metal deposition

DED/LMD feeds powder or wire into a moving melt pool. Its marine significance is twofold. First, it enables *cladding and repair*: HEA overlays on marine steel substrates or worn bronze/steel components deliver corrosion- and wear-resistant surfaces at a fraction of a monolithic HEA part’s cost; laser-clad AlCoCr_xFeNi coatings on steel have been studied specifically in this light [14]. Second, DED supports *in-situ alloying and compositionally graded deposits*, which researchers have used

both to screen compositions and to functionally grade from a steel substrate to a corrosion-resistant HEA surface [13]. Within this route, Ti- or Cu-doped AlCoCrFeNi deposits and Nb–Mo–Ta–W compositions designed for marine service have been reported; the former showed improved corrosion resistance, while the latter—despite better overall performance than 316L—exhibited localized pitting associated with minor compositional segregation, illustrating the defect sensitivity discussed in Section V [13]. Control of dilution and interfacial galvanic behavior is the key engineering issue for clad components.

C. Wire arc additive manufacturing

WAAM deposits material by arc welding of wire feedstock at deposition rates and build volumes far beyond powder-bed processes, which is why it is the route of record for large marine and offshore structures [15]. For HEAs the bottleneck has been wire availability, and three strategies have emerged: cable-type twisted wires (demonstrated for MoNbTaWTi, yielding single-phase BCC deposits with hardness near 533 HV and compressive strength stable between 500 and 1000 °C [31]), combined-cable wire for Al–Co–Cr–Fe–Ni [32], and metal powder-cored wires, now demonstrated for Cantor-type CoCrFeMnNi walls with homogeneous element distribution and competitive mechanical properties [33]. WAAM HEA deposits are coarse and columnar, with cyclic reheating from successive passes; Mn evaporation, oxide and sulfide inclusions, and mild elemental segregation (notably Mn) are recurring metallurgical issues [15], [33]. Because WAAM parts are typically machined to finish, the as-deposited corrosion performance is less critical than the properties of the final machined surface—but inclusions and lack-of-fusion defects remain potential pit initiation sites.

D. Binder jetting and other routes

Binder jetting followed by sintering offers high build speed and low cost but leaves residual porosity unless infiltration or hot isostatic pressing (HIP) is applied; porous binder-jet HEA structures have been proposed for lightweight or functional applications [13]. For load-bearing, seawater-wetted marine parts, residual porosity is a corrosion liability (Section V), so binder jetting is currently a secondary route for this application domain.

E. Feedstock: the binding constraint

Marine deployment is ultimately limited less by process physics than by feedstock economics. Gas-atomized pre-alloyed HEA powders are expensive, and the critical-raw-material content (Co, Ni, Ta, Nb) of many attractive compositions raises both cost and supply-risk questions [12], [13]. Wire production for WAAM—drawing, cabling, or powder-coring—is immature for most HEA compositions, and element evaporation (Mn, Cr) during wire manufacture and deposition must be compensated in the chemistry [33]. Powder reuse and oxidation control in the humid, saline atmosphere of shipyard environments add further qualification requirements. A credible marine HEA program therefore starts with a feedstock strategy, not a printer.

IV. AS-BUILT MICROSTRUCTURE AND MECHANICAL PERFORMANCE

A. Characteristic AM microstructures of HEAs

Across powder-bed processes, as-built HEAs display fine cellular or dendritic solidification substructures, high dislocation densities arranged in cell walls, and epitaxially grown columnar grains aligned with the build direction; texture, cell size, and phase balance are all tunable through laser power, scan speed, hatch spacing, and scan strategy [12], [13]. In single-phase FCC alloys such as CoCrFeMnNi, LPBF produces a homogeneous solid solution with grain sizes well below cast equivalents, which raises yield strength via the Hall–Petch effect while retaining ductility [30]. In eutectic AlCoCrFeNi_{2.1}, SLM produces a fine, relatively homogeneous FCC/B2 duplex structure whose inter-phase composition spread is far smaller than in cast or annealed material [11]. WAAM deposits are coarser and more anisotropic: elongated columnar grains, dendritic substructure, and reheated interpass regions [31], [33].

B. Defects and their engineering meaning

The defect population of AM HEAs mirrors that of AM metals generally: lack-of-fusion pores, gas porosity, solidification cracks, and inclusions, plus HEA-specific risks such as hot cracking in Al-rich compositions and elemental loss by evaporation [13]. For marine service the important point is that every pore or crack intersecting a wetted surface is a candidate pit or crevice initiation site, and internal porosity can connect into crevice-like networks after surface machining [2]. Hot isostatic pressing closes internal porosity and is now standard practice for qualifying powder-bed parts; HIPed LPBF CoCrFeMnNi retains essentially the same chloride corrosion performance as the as-built state while improving density [30].

C. Post-processing: the heat-treatment trade-off

Heat treatment of AM HEAs is a genuine design decision with mechanical benefits and corrosion costs. Homogenization anneals relieve residual stress and coarsen the microstructure, but they can also restore the elemental segregation that rapid solidification suppressed. The clearest evidence comes from SLM AlCoCrFeNi_{2.1}: the as-built alloy, with its homogeneous inter-phase chemistry, shows superior corrosion resistance in 3.5 wt% NaCl, while annealing dramatically degrades passivity as FCC/B2 segregation develops [11]. Similar homogenization–corrosion interactions are reported for Al_xCoCrFeNi systems [12]. Conversely, targeted annealing of LDED AlMo_{0.25}FeCoCrNi_{2.1} has been used to tune strength and corrosion jointly [34]. The design rule: any post-build heat treatment must be qualified for corrosion, not just mechanics.

D. Representative mechanical performance

A few results anchor the current state of the art. LPBF CoCrFeMnNi achieves strength above the wrought baseline with retained ductility, and exhibits notable abrasion resistance—the first low-stress abrasion data reported for any AM HEA [30]. Mo-alloyed EHEAs reach yield strengths near 926 MPa

with compressive strains approaching 40% [24]. WAAM MoNbTaWTi deposits remain hard (around 533 HV) and maintain compressive strength from 500 to 1000 °C, positioning them for hot, abrasive niches such as exhaust-side marine hardware [31]. WAAM Cantor-type walls deliver competitive tensile behavior with anisotropy tied to the columnar grain structure [33]. Finally, hydrogen permeation through LPBF CoCrFeMnNi is markedly lower than through 316L despite a higher diffusion coefficient, a favorable sign for hydrogen uptake and embrittlement resistance in cathodically protected seawater systems [30].

V. CORROSION PERFORMANCE IN SALINE AND SEAWATER ENVIRONMENTS

A. Test environments and methods

Almost all published AM-HEA corrosion data come from laboratory saline media—most commonly 3.5 wt% NaCl or artificial seawater prepared to ASTM D1141 [35]—using potentiodynamic polarization and electrochemical impedance spectroscopy under the conventions of ASTM G3/G5/G61 [36]–[38]. These tests rank alloys and passive-film quality well, but they are not seawater service: they omit biofilms, tidal wet–dry cycling, crevice geometries, flow, and multi-month timescales. Data from natural marine exposure exist for conventional AM alloys (e.g., SLM AlSi10Mg immersed in the Adriatic for 3–6 months [39]), but equivalent long-term exposures of AM HEAs are, to our knowledge, absent from the literature. This is the single most important evidence gap in the field.

B. AM versus cast: the homogeneity advantage

The consistent finding across recent studies is that AM improves the chloride corrosion performance of single-phase and near-single-phase HEAs relative to cast counterparts. Mechanisms are twofold: (i) homogeneous elemental distribution removes micro-galvanic couples between segregated phases, and (ii) lower micro-porosity and finer structure reduce pit initiation sites. Reported results include LPBF CoCrFeMnNi with higher pitting resistance and polarization resistance than cast material [12], and SLM CoCrFeMnNi whose corrosion current density drops by roughly 58% relative to the as-cast alloy, attributed to a more uniform composition and a more stable passive film [12], [13]. In AlCoCrFeNi_{2.1} EHEA, the as-built SLM condition outperforms both annealed and cast specimens in 3.5 wt% NaCl for exactly these reasons [11]. The caveat, again, is that these advantages are hostage to defect control: the same reviews flag that secondary phases, intermetallics, and segregation introduced by poor parameters or heat treatment can create localized corrosion sites [13]. The picture is not uniform: where SLM builds carried higher densities of pores, cracks, and dislocations, the *cast* counterpart outperformed printed CoCrFeMnNi in 3.5 wt% NaCl [40]—build quality, not chemistry alone, sets the ranking (Section IX-B).

C. AM HEAs against the marine benchmark: 316L

The most direct marine-relevant comparison is LPBF CoCrFeMnNi against LPBF 316L. In chloride solution, as-built and

HIPed CoCrFeMnNi exhibit roughly 2.2 times better corrosion performance than 316L, with stable pseudo-passive behavior and higher pitting potentials. In acidic solution the ranking inverts completely: 316L outperforms the HEA by more than an order of magnitude [30]. The lesson for marine specification is that HEA superiority in seawater is real but *environment-specific*; acidic, sulfide-rich, or creviced microenvironments (under deposits, inside crevices, beneath biofilms) cannot be assumed benign on the strength of neutral NaCl tests alone [9].

D. Composition strategies for seawater passivity

Several alloying strategies are emerging specifically for chloride service. Mo additions to EHEAs ($\text{AlCoCrFeNi}_{2.1}\text{Mo}_x$) simultaneously raise strength and pitting potential, with in-situ Mo-oxide/Cr-oxide composite films stabilizing passivity [24]. Ti or Cu doping of LMD AlCoCrFeNi improves corrosion resistance over the undoped alloy, though secondary intermetallics must be suppressed [13]. Zr-containing HEAs have shown improved resistance over Zr-free variants [13], and Nb–Mo–Ta–W refractory compositions designed for marine use beat 316L overall, albeit with deep pits nucleating at minor segregates in one study [13]. CoCrW-type alloys deposited by LPBF show passivation that evolves over weeks of immersion, with corrosion concentrated at melt-pool boundaries—a reminder that AM’s own solidification pattern becomes part of the corrosion microstructure [41].

E. Anisotropy, surface condition, and coupled degradation

Three second-order effects are known from AM metals generally and are only beginning to be characterized in HEAs. *Build orientation* changes the density of melt-pool boundaries and cell structures exposed at a surface, and corrosion rates in LPBF alloys vary with orientation [12], [13]. *Surface condition* dominates localized attack: for LPBF 316L, as-built rough surfaces are markedly more SCC-susceptible than machined ones, and cracks propagate along melt-pool boundaries [3]; machined AM surfaces can even exceed wrought behavior in pitting tests [2]. *Coupled degradation*—corrosion fatigue, SCC, erosion-corrosion, and cavitation—is where marine components actually fail, yet AM-HEA data in these modes are essentially nonexistent; hydrogen permeation evidence for LPBF CoCrFeMnNi is one of the few positive indicators [30].

F. Galvanic and biofouling interactions

For DED-clad or WAAM-on-steel components, galvanic coupling between the HEA deposit and the substrate (or between an AM HEA part and adjacent conventional alloys) must be assessed; passive HEAs are typically noble relative to structural steel, so coating defects concentrate attack on the substrate side, and compatibility with cathodic protection regimes (e.g., offshore CP design practice) needs verification [2]. Biofouling interactions with HEA surfaces—which couple biological films to under-film chemistry and may accelerate localized attack—have not yet been studied in any systematic way. Table X summarizes the key corrosion evidence base.

VI. MARINE APPLICATIONS AND DEPLOYMENT PATHWAYS

A. Where AM HEAs fit in the marine system

The argument for AM HEAs is strongest where corrosion cost, geometric complexity, or repair difficulty are simultaneously high, so that the material premium is amortized. Table XI maps candidate applications; the main clusters are:

Seawater systems and pumps. Impellers, valve trim, strainers, and fittings in ballast, cooling, and fire-main systems are small, complex, and corrosion-critical—an ideal LPBF niche, echoing LPBF’s established role in complex high-performance components [13]. FCC CoCrFeMnNi-type alloys and Mo-modified EHEAs are the leading composition candidates given their chloride performance [24], [30].

Heat exchangers and desalination hardware. LPBF’s ability to print conformal internal channels enables compact seawater heat exchangers; the same geometrical freedom applies to energy-recovery and reverse-osmosis components, where pitting and crevice resistance govern life [2], [13].

Large structures and propulsion. WAAM suits propeller blanks, brackets, offshore nodal repairs, and other meter-scale items where powder-bed volumes are impractical [15]; the economics depend on HEA wire becoming available at scale [33].

Repair and cladding. DED overlays of HEA on marine steel, or on worn bronze and steel components (shafts, seal seats, rudder stocks), deliver corrosion- and wear-resistant surfaces only where needed [14]. This is arguably the fastest route to impact because it sidesteps whole-part cost and leverages existing shipyard welding infrastructure.

B. Hybrid and functionally graded strategies

Because feedstock cost dominates, the near-term deployment model is hybrid: conventional marine steel for structure, AM HEA where the seawater actually acts. Compositionally graded DED builds—steel to HEA transition—can manage both cost and galvanic compatibility, provided dilution and intermetallic formation at the interface are controlled [13]. Cathodic protection compatibility must be checked case by case, since passive HEAs shift the galvanic balance of a protected assembly [2].

C. Qualification: what classification societies need

Marine deployment requires classification-society acceptance. DNV’s standard for additive manufacturing of metallic parts (DNV-ST-B203 [42]), ABS’s metal AM guidelines [43], and Lloyd’s Register’s AM requirements [44] currently provide frameworks developed around conventional engineering alloys; HEA compositions sit outside the qualified material list and would need to be qualified as new materials, with mechanical and corrosion datasets. A defensible qualification ladder for an AM HEA marine component is:

- 1) *Material baseline:* composition control, powder/wire certification, defect acceptance criteria (porosity, cracks, inclusions) with NDT.
- 2) *Electrochemical screening:* polarization and EIS in ASTM D1141 seawater [35] under ASTM G3/G5/G61

conventions [36]–[38], benchmarked against the incumbent alloy (e.g., 316L, NiAl bronze) in identical conditions.

- 3) *Long-term exposure*: immersion and wet–dry cycling in natural or artificial seawater for months, with post-exposure mechanical re-testing, mirroring exposure protocols already used for conventional AM alloys [39].
- 4) *Coupled degradation*: corrosion fatigue and SCC (e.g., slow-strain or U-bend tests in seawater), erosion-corrosion and cavitation for rotating parts [2], [3].
- 5) *System level*: galvanic compatibility tests in the actual assembly, CP compatibility, and pilot installation with inspection intervals.

Table XII maps the applicable standards onto this ladder. The gap worth emphasizing: no existing marine AM standard references HEA compositions, and no HEA standard references marine exposure classes—closing that two-way gap is the qualification task. The Data Appendix (Section IX) turns each rung of this ladder into quantitative detail: electrochemical synthesis with 316L benchmarks, defect–pit initiation statistics and acceptance criteria, galvanic/CP protocols, a minimal natural-seawater exposure matrix, and an expanded test-method mapping.

VII. CHALLENGES AND OUTLOOK

A. Open technical challenges

Feedstock economics and supply. Powder costs and critical-element exposure (Co, Ni, Ta, Nb) remain the largest barrier to marine adoption; WAAM wire routes mitigate cost but introduce evaporation and inclusion control problems [13], [33]. Development of low-Co or Co-free corrosion-resistant HEAs is strategically important.

Defect–corrosion linkage. The field lacks quantitative defect-tolerance criteria: how much porosity, of what size and connectivity, can a seawater-wetted AM HEA surface tolerate before pitting statistics become unacceptable? Such criteria exist implicitly for aerospace fatigue but not for marine corrosion, and they will be needed for classification [42].

Environment-specificity of passivity. The 316L-comparison evidence shows HEA passivity can collapse outside neutral chloride media [30]. Crevice chemistry, under-biofilm acidification, and sulfides in stagnant zones all push toward those regimes and are untested for AM HEAs [9].

Coupled degradation modes. Corrosion fatigue, SCC, cavitation, and erosion-corrosion data for AM HEAs are nearly absent, yet these govern marine component life [3].

Biofouling–corrosion interaction. Whether HEA surfaces, with their distinct passive-film chemistry, resist, tolerate, or accelerate biofilm-mediated attack is simply unknown; this connects the present topic to the companion literature on sustainable biofouling control [2].

B. Research directions

Corrosion-by-design. The demonstrated loop of thermodynamic design plus electrochemical validation [28], [29] should be closed with machine-learning surrogate models trained on AM-specific microstructures, targeting compositions that are

simultaneously printable (hot-crack resistant) and passive in seawater.

Process–passivity maps. Systematic studies linking LPBF/WAAM parameter windows to passive-film composition (e.g., Cr/Mo enrichment measured by surface analysis) would convert corrosion from a screening outcome into a process-controlled property [11], [41].

Real-seawater exposure networks. Long-term immersion programs—tidal, splash, and submerged zones, with and without biofilm—for a small matrix of benchmark AM HEAs versus incumbent alloys, following the natural-exposure template already demonstrated for AM aluminum [39].

Hybrid structures. Functionally graded DED builds and HEA claddings on marine steel, with galvanic and CP compatibility designed in, are the most economical near-term pathway [13], [14].

Digital qualification. In-situ melt-pool monitoring plus build-data archives could support the data-driven material qualification frameworks classification societies are moving toward [42], [43], shortening the path for each new HEA composition.

C. Deployment roadmap

A realistic staged roadmap is: (1) *now*—benchmarking studies and repair cladding pilots on non-class-critical hardware; (2) *near term* (2–5 years)—qualified LPBF small components in seawater systems, WAAM propeller and bracket demonstrations, and the first natural-seawater exposure datasets; (3) *medium term* (5–10 years)—class-approved HEA compositions within marine AM standards, functionally graded marine structures, and life-cycle-cost validated substitution of incumbent alloys. Sustainability assessment should run alongside: HEA-AM parts must justify their critical-element burden through longer life, less maintenance, and recyclability to be credible in an industry moving toward life-cycle accountability [2].

VIII. CONCLUSIONS

- 1) HEAs with FCC CoCrFeMnNi-type or Mo-modified eutectic compositions exhibit passive-film corrosion resistance in neutral chloride media that matches or exceeds the 316L benchmark, making them credible candidates for seawater-wetted marine components [9], [24], [30].
- 2) Additive manufacturing is not merely a shaping tool but a *corrosion-engineering* tool for HEAs: rapid solidification suppresses elemental segregation, refines microstructure, and measurably improves pitting resistance over cast material—provided defects and post-build segregation are controlled [11]–[13].
- 3) Process selection maps onto marine needs: LPBF for complex small components in seawater systems, SEBM for crack-prone and refractory systems, DED for cladding, repair, and functionally graded construction, and WAAM for large structures once HEA wire feedstock matures [13]–[15].
- 4) Corrosion performance is environment-specific: AM CoCrFeMnNi surpasses 316L in chloride but is far

inferior in acidic media, so crevice, under-deposit, and biofilm microenvironments must be tested explicitly rather than extrapolated from NaCl screening [30].

- 5) The decisive gaps for deployment are evidential and regulatory: long-term natural seawater exposure, corrosion fatigue/SCC, biofilm coupling, and galvanic/CP compatibility data are scarce, and no marine AM standard yet accommodates HEA compositions [42]–[44].
- 6) The practical near-term pathway is hybrid: HEA claddings and graded deposits on marine steel, qualified through a staged ladder of electrochemical screening, long-term exposure, coupled-degradation testing, and system-level galvanic assessment.

IX. DATA APPENDIX: QUANTITATIVE SYNTHESIS AND QUALIFICATION PROTOCOLS

This appendix consolidates the quantitative evidence behind Sections V–VI and specifies the test protocols a qualification program would actually run. Where the literature is heterogeneous, we report ranges and state the reasons formal meta-analysis is not yet possible. *Flagging convention:* items marked [E] are evidence-based statements anchored to published data or standards text; items marked [P] are proposals, estimates, or assumptions of this work that require experimental validation. A machine-readable version of the electrochemical synthesis (Table III) with source identifiers and reference-electrode normalization notes is provided in `data/electrochem_quant.csv`.

A. A1. Quantitative electrochemical synthesis vs. LPBF 316L

Table III compiles the principal electrochemical data for the main AM-HEA families in chloride media, with 316L benchmarks. Three cautions govern interpretation. *First*, reference electrodes differ between studies (SCE, Ag/AgCl, saturated calomel); values are quoted as published and can shift by tens of mV when converted, so cross-study comparisons of absolute potentials carry an uncertainty of that order. *Second*, surface finish and sweep protocol are not uniform; reported i_{corr} for the same alloy family spans roughly an order of magnitude across studies. *Third*, build quality is a hidden variable: the same nominal alloy can rank above or below cast material depending on defect content (Section IX-B). Consequently, honest error bars cannot be assigned today; the ranges in Table III are the defensible summary statistic.

The key magnitudes: corrosion current densities of well-built CoCrFeMnNi-type deposits cluster around $0.09\text{--}0.13\ \mu\text{A}/\text{cm}^2$ in aerated 3.5 wt% NaCl, with pitting potentials of order +0.1 to +0.2 V versus common references [45]; one SLM study reports $E_{\text{pit}} = 197\ \text{mV}$ for AM against only 37 mV for cast, with passivation windows of 386 vs. 216 mV [45]. For the LPBF 316L reference itself, AM material typically *exceeds* wrought: critical pitting temperature of $27.5\text{--}31\ ^\circ\text{C}$ (build-direction dependent) against $16\ ^\circ\text{C}$ for wrought in 1 M NaCl, and pitting potentials some 330 mV nobler at $25\ ^\circ\text{C}$ [46]. This sets a high bar: an AM HEA must beat a strong AM-316L baseline, not just a wrought one. EHEAs passivate readily in neutral NaCl (AlCoCrFeNi_{2.1} between 5 and $45\ ^\circ\text{C}$), but lose

passivity completely in the presence of reducing bisulfite—a proxy for occluded/crevice chemistry [47]—and Mo additions lift the pitting potential to +282 mV with low passive current [24]. Charge-transfer resistances for passive CoCrFeMnNi-type surfaces are of order $10^5\ \Omega\cdot\text{cm}^2$, comparable to 304 SS [48].

A minimum reporting standard for future work should follow: potentials converted to and reported against Ag/AgCl/seawater; surface preparation stated per ASTM G1 [49] (e.g., 600-grit SiC finish, ultrasonic clean); sweep conventions per ASTM G3/G5 [36], [37]; $n \geq 3$ specimens per condition with spread reported; and reference alloys (LPBF 316L, NiAl bronze) tested in the same cell.

B. A2. The LPBF CoCrFeMnNi vs. LPBF 316L benchmark: conditions and robustness

The widely quoted “ $\sim 2.2\times$ better in chloride, $\sim 15.5\times$ worse in acid” result comes from a single study in the journal *High Entropy Alloys & Materials* [30]. From the published record: specimens were printed in two orientations (perpendicular and parallel to the build direction) at an optimized volumetric energy density (window $60\text{--}110\ \text{J}/\text{mm}^3$, optimum $104\ \text{J}/\text{mm}^3$); porosity was quantified by X-ray micro-CT; both as-built and HIPed conditions were tested; media were 3.5 wt% NaCl and 1 N H₂SO₄, with 316L and 430 SS as references; HIPed material retained the chloride advantage while hardness fell from 229 ± 5 to $152 \pm 4\ \text{HV}$ [30]. Surface preparation for the electrochemistry is not fully documented in the open record—a gap reviewers should note; standard practice is grinding to at least 600-grit per ASTM G3/G5 conventions [36], [37].

Robustness is not yet demonstrated. No multi-lot powder or multi-build replication has been published for this benchmark, and the wider literature contains direct contradictions: one SLM-vs-cast study found the *cast* alloy more resistant in 3.5 wt% NaCl because its SLM builds carried higher densities of pores, cracks and dislocations (i_{corr} : cast 1.20×10^{-7} vs. XY 1.23×10^{-7} vs. XZ $1.33 \times 10^{-7}\ \text{A}/\text{cm}^2$; E_{corr} vs. SCE: -0.264 , -0.346 , $-0.323\ \text{V}$ respectively) [40], while SLM CoCrNi has likewise been reported inferior to cast in one study and superior in another [50]. The reconciling variable is defect content, not chemistry alone (Section IX-C). Build-orientation effects on $E_{\text{corr}}/i_{\text{corr}}$ of a few tens of mV and $\sim 10\text{--}20\%$ are consistently observed [40], [51]. A qualification-grade version of the benchmark therefore requires: ≥ 2 powder lots, ≥ 3 builds with different orientations per condition, documented surface preparation, and both NaCl and crevice-type media.

Provenance of the “ $\sim 2.2\times$ ” claim—what is and is not in the record [E]: the statement derives from comparative potentiodynamic polarization of AB and HIPed CoCrFeMnNi against LPBF 316L (and 430 SS) in 3.5 wt% NaCl and 1 N H₂SO₄ [30]. In the publicly accessible record: (i) the metric underlying the ratio is the comparative polarization response in chloride (corrosion performance as assessed from the polarization curves); the absolute E_{pit} , CPT, polarization resistance and i_{corr} values, specimen counts (n) and standard deviations are *not* reproduced in the open abstract/full-text surfaces we could verify, so the ratio cannot yet be decomposed into standardized metrics; (ii) surface preparation is documented

only as the as-built and HIPed conditions, with electrochemical surface finishing not fully specified; (iii) specimen counts and SD are not reported. [P] We therefore treat the $2.2\times$ figure as a directional result and recommend: a data request to the authors for the underlying curves with n and SD; conversion of all potentials to Ag/AgCl/seawater using the CSV conversion convention (SCE $\rightarrow$ Ag/AgCl-seawater: -9 mV); and the interlaboratory protocol of Section IX-K before the ratio is used in any specification. Table III (row 5) and `data/electrochem_quant.csv` flag the ratio explicitly as *measured_ratio* with NA in the normalization columns.

C. A3. Defect classes and pit initiation statistics

Quantitative defect–corrosion links exist mainly from AM stainless steels and transfer to HEAs by analogy (Table IV). The strongest anchors: (i) LPBF 420 SS with lack-of-fusion (LOF) pores averaging $18\text{ }\mu\text{m}$ at 0.42% porosity developed $\sim 100\text{ }\mu\text{m}$ pits at defect zones versus $<2\text{ }\mu\text{m}$ pits in defect-free zones—a roughly $50\times$ pit-size amplification from a few tenths of a percent of porosity [52]; (ii) for SLM 316L, total porosity in the 0.01–0.4% range did not shift the pitting potential but increased metastable pitting and lowered the repassivation potential, i.e., porosity governs initiation statistics and stability of propagation rather than the thermodynamic pitting threshold [53]; (iii) minimizing LOF in DED 304L significantly improved global corrosion response [53]; and (iv) LOF pores are structurally complex—oxide-film-lined cavities with nanocrystalline shells, predominantly spatter-derived—so each surface-breaking LOF is a pre-formed crevice with occluded chemistry [54]. MnS-type inclusions, the classic pit initiators in wrought stainless, are refined or absent in LPBF material, which is why even rough as-built LPBF 316L often outperforms wrought [55]; in WAAM HEAs, by contrast, MnS and oxide inclusions have been observed and must be counted [33]. Location effects (build-plate position relative to shielding-gas flow) measurably change pit initiation and should be mapped [53].

For defect acceptance criteria we recommend adapting fatigue-style criteria to corrosion service: (a) near-surface ($<200\text{ }\mu\text{m}$ depth) LOF classified separately from interior porosity, since only surface-connected defects initiate external pitting; (b) maximum-pore-size limits rather than volume-fraction-only limits, given the pit-amplification evidence [52]; (c) inclusion density limits for WAAM deposits; and (d) statistical validation using the small-electrode method—measuring initiation-site density as a function of electrode area, as demonstrated for sintered 316L—to obtain pit-initiation *statistics* rather than single-curve pass/fail outcomes [56].

D. A4. Galvanic protocols and cathodic protection compatibility

For cladding and repair applications, galvanic behavior must be tested, not inferred. Recommended protocol per ASTM G71 [57]: duplicate or triplicate HEA–substrate couples (HEA on AH36/marine steel; HEA on NiAl bronze for propeller repair) in ASTM D1141 seawater [35] and, where possible, natural seawater; area ratios spanning the service extremes (small HEA repair on large steel structure $\Rightarrow$ large cathode/small anode ratio

reversed and vice versa); continuous current logging for 30–90 days plus mass-loss validation; and free-corrosion potential mapping of each material before coupling.

Acceptance thresholds: no universal numerical criterion exists for HEA couples, so we propose a two-part engineering criterion: (i) the galvanic current density shall not raise the substrate’s corrosion rate by more than 10% of its uncoupled value (or below the design corrosion allowance rate, whichever is stricter); and (ii) no localized attack shall initiate at the HEA/substrate interface or at coating defects during the test. Passive FCC HEAs are expected to be noble to structural steel, so the protected member is the substrate at defects—exactly the geometry where acceptance matters.

Interaction with cathodic protection (CP): offshore CP design typically holds carbon/low-alloy steel at or below about -0.80 V vs. Ag/AgCl/seawater with sacrificial anodes, and limits potentials more negative than roughly -1.10 V vs. Ag/AgCl to avoid hydrogen damage to higher-strength components [58]. The qualification questions for an HEA in a CP system are therefore: does the HEA remain passive (or behave acceptably) at -0.80 V imposed potential; does it inflate the CP current demand of the assembly; and, for any martensitic or high-strength HEA regions (e.g., in graded deposits), is hydrogen uptake acceptable—one LPBF CoCrFeMnNi study reports hydrogen permeation about five times lower than 316L, a favorable indicator [30], but imposed-potential testing in D1141 at the CP set point is required evidence.

E. A5. Minimum natural-seawater exposure matrix

ASTM G52 provides the governing practice for surface-seawater exposure, covering full immersion, tidal and splash zones, and spray, with crevice configurations per Guide G78 and control materials per Test Method D3623 [59], [60]. Table V specifies a minimal matrix we judge persuasive for an initial class-society submission. Design principles: at least two exposure sites with different temperature/salinity/biota; triplicate specimens per cell; two surface conditions (as-built and machined/polished) and two build orientations for AM material; crevice assemblies on every retrieval interval; and reference alloys (LPBF and wrought 316L, NiAl bronze, 2205) in identical racks. Durations of 1, 6, 12 months are the persuasive minimum, with 24-month follow-ups for any component seeking full class approval. Endpoints per interval: mass loss per ASTM G1/G31 [49], [61], maximum pit depth and rating per ASTM G46 [62], retained tensile properties, crevice inspection, and biofilm sampling (Section IX-I). Note that G52 explicitly excludes high-velocity service; pumped-seawater velocity loops (0.5–3 m/s) must be added separately for heat-exchanger and pump qualification.

F. A6. Integrating computational screening with printability

The components of a closed-loop design pipeline now all exist. *Printability*: a demonstrated high-throughput framework couples CALPHAD thermo-physical properties (solidus/liquidus, Scheil paths) with fast melt-pool surrogate models and physics-based defect criteria for lack-of-fusion, balling, keyholing and hot cracking, producing probabilistic

printability maps; for the equiatomic CoCrFeMnNi space the defect-free processing fraction (printability index) was 17–22%, and Co–Fe- and Cr–Co-rich sub-regions offered the best trade-off between balling and hot-crack resistance [63]. Hot-crack susceptibility is quantified by Kou’s crack susceptibility index (maximum of $|dT/df_s^{1/2}|$ near the terminal solidification stage) computed from CALPHAD Scheil paths, an approach validated by weldability data for refractory systems and extendable to HEA databases [64]. *Passivity*: composition-to-corrosion surrogates can be trained on the growing polarization datasets for HEAs [9], [10], augmented by DFT-derived dissolution and oxide-stability descriptors and by SKPFM work-function contrasts between phases as micro-galvanic proxies [28], [29]. *Integration*: multi-objective Bayesian or genetic-algorithm optimization over (printability index, CSI, passivity proxy, cost/critical-element penalty) produces a Pareto front of candidate compositions; hybrid ML–CALPHAD frameworks are the current best practice for such coupled objectives [65]. *Validation*: compositionally graded DED libraries or LPBF combinatorial coupons test the top candidates in days, feeding back to the surrogates. WAAM screening differs mainly in feedstock constraints (wire manufacturability, evaporation loss per Section IX-H) entering the objective function.

G. A7. Post-processing balance for EHEAs

Table VI summarizes the evidence. The controlling fact for EHEAs is the heat-treatment trade-off: SLM AlCoCrFeNi_{2.1} in the as-built state has the most homogeneous inter-phase chemistry and the best chloride corrosion resistance, while annealing restores FCC/B2 segregation and dramatically degrades passivity [11]. HIP is the safest densification route because it closes porosity without deliberately driving segregation: HIPed LPBF CoCrFeMnNi kept chloride performance equal to as-built [30], at the cost of softening (229 → 152 HV). For surface finishing of wetted faces, 316L analogues are strong: CW laser polishing (remelting mode) cut roughness by 85% (Sa 2.1 → 0.3 μm), reduced i_{corr} by 97% and widened the passive window by 98%, whereas nanosecond polishing introduced microcracks and femtosecond polishing gave limited benefit [66], [67]; electropolishing reaches internal channels, enriches the passive film in Cr (Cr/Fe ratio roughly tripled), and cut i_{corr} from 2.5 to 0.8 $\mu\text{A}/\text{cm}^2$ in 316L [68]. Recommended EHEA sequence: HIP → machining to near-final geometry → electropolish or CW laser polish of wetted surfaces → low-temperature stress relief only if required (and corrosion re-qualified). Any homogenization anneal must be justified mechanically and re-qualified electrochemically.

H. A8. Mn/Cr evaporation in WAAM: magnitude and compensation

Evaporation is a real but manageable effect. In cable-wire WAAM of CoCrFeMnNi with a pre-calculated chemistry, measured deviations were modest for the principal elements (Mn ~2%, Ni ~10% relative error) but large for minor, volatile, or surface-active additions (Al, Si) [69]. Metal-cored-wire WAAM of a CoCrFeMoNiV alloy showed no clear bulk burn-off, with only slight losses of Cr, Mo and Co [70];

Cantor-type MPCW deposits exhibit minor Mn segregation and MnS/oxide inclusions [33]. Practice guidance: (i) over-alloy Mn in the wire (welding-wire practice uses 1–3 wt% excess) and *verify deposited chemistry per build*—deposited composition is a process output, not an inference from wire chemistry; (ii) prefer low-heat-input transfer modes (CMT/short-circuit) with shorter molten-pool residence time; (iii) use trailing and side shielding or local enclosures—in reactive-alloy WAAM, supplementary shielding cut surface oxygen from ~32 to ~6 wt%, illustrating the magnitude of atmosphere control available [71]; (iv) keep Cr above the passivity threshold (order 15 at%) after losses, since Cr supply to the passive film is the corrosion-critical variable. Corrosion-wise, mild Mn depletion is not automatically harmful—the absence of MnS inclusions is what lifts pitting resistance in LPBF 316L [55]—but it must be measured, not assumed.

I. A9. Biofilm formation and under-film chemistry on HEAs

Preliminary studies *do* exist—this is no longer a blank field, though no AM-HEA biofilm study in natural seawater has been published yet. First-generation results: marine *Pseudomonas aeruginosa* biofilms on FeCoCrNiMo_{0.1} thinned and weakened the passive film and drove selective Fe dissolution via extracellular electron transfer (the first HEA MIC study) [18]; on Cantor-type FeCoNiCrMn in simulated seawater, the same organism produced maximum pit depths of 4.77 μm at 14 days against 3.16 μm sterile (~1.5 $\times$), with OCP dropping from –367 to ~–500 mV and polarization resistance falling [19]; and the sulfate-reducing bacterium *Desulfovibrio vulgaris* caused sulfide formation, passive-film thinning and pitting on CoCr–FeMnNi, becoming *more* aggressive under carbon-starvation conditions that upregulate hydrogenase-mediated electron uptake [20]. The practical implication: sterile-solution screening underestimates localized attack, and occluded, nutrient-limited niches are the aggressive case.

Table VII outlines a program to close the gap for AM HEAs.

First-line qualification organism set and acceptance endpoints [E]/[P]: beyond *P. aeruginosa* and *D. vulgaris*, a defensible first-line set is: a marine Gram-negative colonizer (*Vibrio* or *Pseudoalteromonas* sp.), a sulfur-oxidizing acidifier (*Acidithiobacillus thiooxidans*-type), and a natural-seawater mixed-biofilm control (field rack coupons, per the 16-week microfouling chronology established for CuNi/Ti with ASTM G1 preparation [23]). Copper-containing HEA variants (Fe_{0.75}Cu_{0.25}CoNiCrMn: 89.5% antibacterial rate at 14 days) should be screened in parallel as a mitigation strategy [21], [22]. Exposure durations: 7, 14 and 28 days laboratory (bracketing the 14-day standard used in all HEA MIC studies to date [18]–[20]), plus 16-week natural exposure. Monitoring and enumeration follow NACE TM0194 (field monitoring of bacterial growth, sessile counts) and NACE TM0212 (detection, testing and evaluation of MIC) conventions [72]–[74]. Proposed acceptance endpoints [P] (aligned with ASTM practice where possible): (i) maximum pit depth ratio (biotic/sterile) ≤ 1.5 , anchored to the measured 1.51 $\times$ Cantor-alloy result [19]; (ii) polarization resistance retention $\geq 70\%$ of sterile control at 14 days; (iii) no breakdown potential below the sterile control

value; (iv) pit density rating per ASTM G46 [62]; (v) sessile cell counts within one order of magnitude across replicates per TM0194. Note the CP–MIC interaction warning: on pipeline steel, -850 mV vs. SCE CP was insufficient to suppress *D. vulgaris* MIC and even favored biofilm development [74]—HEA qualification must include an inoculated arm at imposed CP potential, not just free corrosion.

J. A10. Expanded standards mapping

Table XII now includes the pitting/crevice (ASTM G48, G78, G192, G46), stress-corrosion (G30, G38/G39, G129, G123), galvanic (G71), and field-exposure (G52, G31) methods proposed for the qualification ladder of Section VI, together with the CP design reference [58].

K. A11. Robustness of the LPBF-HEA/316L benchmark: assessment and interlaboratory protocol

Assessment [E]: the chloride advantage and acid inferiority of LPBF CoCrFeMnNi relative to LPBF 316L currently rest on a single published benchmark with one powder lot, one optimized parameter window, and surface preparation not fully documented in the open record [30]. Independent replication does not exist; contradictory results exist for the AM-vs-cast comparison [40]. Orientational effects on $E_{\text{corr}}/i_{\text{corr}}$ of tens of mV and ~ 10 – 20% are documented [40], [51], and powder-lot effects (satellite content, oxidation state, flowability shifting melt-pool stability and defect populations) are established drivers of corrosion-relevant defects in AM generally [52]. Conclusion: the benchmark is *hypothesis-grade*, not qualification-grade.

Minimal interlaboratory protocol [P] (round-robin design, ≥ 3 laboratories):

- 1) **Materials:** one CoCrFeMnNi powder specification from two lots (distinct atomization campaigns, documented COA, oxygen content); LPBF 316L control powder, one lot.
- 2) **Builds:** per lab and lot, three builds on different days, each containing tensile + electrochemistry coupons in two orientations (XY, XZ), printed at the lab's own qualified window plus a shared nominal window; density by Archimedes + μ CT near-surface scan; report VED-equivalent metrics.
- 3) **Surface states:** (i) ground to 600-grit SiC per ASTM G1 [49]; (ii) polished to $1\ \mu\text{m}$; both ultrasonically cleaned, dried, tested within 24 h.
- 4) **Electrochemistry:** ASTM D1141 medium [35], aerated and deaerated; OCP for 1 h, then potentiodynamic polarization per ASTM G5 [37] at $0.167\ \text{mV/s}$; EIS at OCP; all potentials reported vs. Ag/AgCl/seawater and converted per the CSV convention; $n \geq 3$ specimens; one additional run in $0.5\ \text{M}\ \text{H}_2\text{SO}_4$ to map the acid-side behavior.
- 5) **Statistics and acceptance:** mixed-effects analysis (lab, powder lot, build, orientation as factors) per ASTM G16 [75]; the benchmark is accepted as robust only if the HEA's lower 95%-confidence E_{pit} across labs exceeds the co-tested 316L median in chloride and the lab-to-lab

variance is reported and bounded. A shipped reference coupon set (same build plate) tested by all labs separates material from laboratory effects.

L. A12. Evidence base for the galvanic acceptance criterion

The $\leq 10\%$ -increase-over-uncoupled acceptance criterion of Section IX-D is a proposal [P], but it is anchored by analogy evidence [E]: (i) DIN 50919-type laboratory couples in synthetic seawater show area ratio, not metallurgy alone, controlling damage—unalloyed steel coupled to 316-type stainless lost $1.00\ \text{mm/yr}$ at 10:1 cathode:anode area ratio but only 0.07 – $0.12\ \text{mm/yr}$ at favorable ratios; a 6%-Mo stainless drove still higher rates, confirming that noble passive cathodes (which is what passive HEAs will be) intensify attack on coupled steel [76]; (ii) the same family of data, inverted, is the basis of Lee–Tuthill-style sacrificial design, where controlled carbon-steel area suppressed 304/316 crevice corrosion up to 100:1 SS:CS ratios at $14\ ^\circ\text{C}$ [77]—demonstrating that couple currents are designable quantities; (iii) for passive-metal/active-metal couples, mixed-potential analysis shows the passive member corrodes near its passive current ($\sim 0.1\ \mu\text{A}/\text{cm}^2$ scale) while the couple current is set by cathodic oxygen reduction on its area [78], which is exactly why HEA area (cathode) drives steel (anode) loss. The 10% value itself is conventional engineering conservatism (keeps the couple within the corrosion-allowance budgeting practice used for design corrosion allowances) rather than a measured threshold [P].

Adaptation under CP [P]: when the assembly is cathodically protected to the DNV-RP-B401 criterion ($\sim -0.80\ \text{V}$ vs. Ag/AgCl/seawater) [58], the free-corrosion couple is replaced by imposed potential: acceptance then becomes (i) the HEA contributes no current-demand increase beyond the CP design reserve (evaluate as added cathodic area with its current density at $-0.80\ \text{V}$, measured per ASTM G71 [57]); (ii) no hydrogen-related damage in any high-strength or martensitic region of the deposit at the most negative operating potential (limit $\sim -1.10\ \text{V}$) [58]; (iii) at coating holidays on adjacent steel, no localized attack escalation beyond the 10% criterion. Tests must therefore be run at three potentials: free couple, $-0.80\ \text{V}$, and $-1.10\ \text{V}$.

M. A13. WAAM over-alloying ranges and chemistry-drift qualification

Measured magnitudes of evaporation: extreme for highly volatile elements (Mg losses up to $\sim 20\ \text{wt}\%$ in Mg-alloy WAAM [71]; Mg losses near one-fifth of wire content at high current and up to $\sim 5.5\%$ absolute across deposition-speed changes in Al WAAM [79]), but modest for Mn/Cr-bearing systems with controlled transfer: metal-cored-wire deposits of CoCrFeMoNiV show no clear bulk burn-off, only slight Cr/Mo/Co losses [70], and pre-compensated cable-wire Cantor builds landed principal elements within a few percent [69]. Filler-wire practice already over-alloys Mn/Si as deoxidizers (e.g., ER70S-6-type wires), and duplex-WAAM work deliberately tunes Mn/N in the wire for phase balance and corrosion [80]. Recommended ranges [P]: for open-arc GMAW/CMT deposition of Cantor-type HEAs, start with +2

to +5 wt% Mn and +1 to +3 wt% Cr over-alloying in the wire, tightening after the first measured wire-to-deposit balance; prefer CMT/low-heat-input transfer and trailing/side shielding or local enclosure, since supplementary shielding demonstrably collapses oxidation (32→6 wt% surface oxygen in a reactive-alloy case [71]); keep deposited Cr above ~15 at% for passivity.

Chemistry-drift qualification in a class environment [P], consistent with DNV-ST-B203 traceability expectations [42]: (i) one full ICP-OES analysis per wire lot (certificate) plus XRF/OES spot checks of the deposit; (ii) PMI at build start, mid-height, and top of each qualification build, with Δ = deposit–wire reported element by element, following the “deposited chemistry is a process output” principle [71]; (iii) drift acceptance bands [P]: $\pm 10\%$ relative for Mn, absolute Cr floor at the passivity threshold, $\pm 5\%$ relative for Co/Ni/Fe; (iv) wire-lot-to-build mapping retained in the build data package, so any field anomaly is traceable to chemistry; (v) requalification on any change of wire supplier, transfer mode, shielding arrangement, or interpass temperature window.

N. A14. Indicative cost model and deployment decision tree

Table VIII gives an indicative first-cost model for a representative seawater pump impeller, and Fig. 3 the decision logic that accompanies it. Anchors [E]: 316L LPBF powder \$20–40/kg (market surveys) with retail lots near \$35–51/kg [81], [82]; LPBF machine rates \$50–150/h, material typically 30–40% of part cost, post-processing 20–35% of total cost [82], [83]. Everything HEA-specific is estimate [P]: HEA powder \$150–400/kg (Co/Ni-rich, small-lot gas atomization), HEA metal-cored wire \$40–120/kg, machine rates for DED/WAAM extrapolated below LPBF, incumbent NiAl bronze casting and its 5–8 yr refurbish/replace interval. No published HEA-AM cost benchmark exists; these orders of magnitude are stated so that they can be replaced by real quotations without changing the decision structure. The economically robust findings are structural, not numeric [P]: (a) cladding/repair converts an HEA decision from a whole-part cost into a surface-area cost; (b) WAAM converts it into a wire-cost decision; (c) only LPBF remains powder-and-machine-hour dominated; (d) all three require evidence-gated maintenance-interval extension (Section IX-E) before they beat cheap incumbents on life-cycle cost, and the LCA must carry the critical-element (Co) burden honestly [2].

O. A15. Crevice geometries and torque protocols for worst-case niches

ASTM G78’s multiple-crevice assembly (MCA) remains the literature-standard geometry even though ASTM has withdrawn the designation (G78-15 marked withdrawn): TFE-fluorocarbon segmented (serrated) washers producing many nominally identical crevice sites per specimen—24 sites per specimen in NRC service implementations, specimen-to-fixture area ratio ~1:3—with serrations existing precisely to give initiation statistics from a few replicates [84], [85]. Torque is the controlling variable: G48 Method D specifies 0.28 N·m and Method F 1.58 N·m for stainless assemblies [85], [86];

the NRC program used 0.70 N·m for metal-to-PTFE (the PTFE limit) and 8.4 N·m for tight metal-to-metal crevices, and reported corrosion appearing only at the higher torque in borderline cases [84]; authors have further reported that 2–8 N·m gives the most reproducible crevice results in neutral chloride service [85]. Recommended AM-HEA protocol [P]: (i) MCA per G78 geometry with PTFE-sleeved fasteners to exclude fixture-metal effects; (ii) two torque levels per coupon set—0.28 N·m (standard screening) and a tightened 2–3 N·m level representing bolted-flange worst case—reporting site-count statistics (number of active sites out of 24) rather than pass/fail; (iii) at least one dissimilar-metal crevice variant (HEA specimen with 316L and with NiAl-bronze serrated washers) to capture galvanic-crevice coupling, mirroring NRC dissimilar-crevice testing [84]; (iv) a metal-to-metal variant (HEA-on-steel shim crevice) at 8.4 N·m to emulate tight structural gaps; (v) complementary electrochemistry by G192 (crevice repassivation potential) and G48-B/F rankings [86], [87]; (vi) 30-day recommended exposure with staged retrievals [60], [85]. For AM material specifically, one MCA set should be mounted on the as-built downskin surface, since its roughness and partially melted powder make it the natural worst-case crevice former [55].

P. A16. Registered proposals: status and promotion criteria

To avoid conflating established practice with this review’s recommendations, all [P] items are consolidated in Table I. *Status declaration*: this review is a literature synthesis; **none of the listed proposals are claimed to be in pilot use by the authors or their partners at submission**. They are hypotheses offered for community discussion and experimental validation; any group piloting one is invited to annotate it at camera-ready or in a commentary, which would be the fastest way to convert [P] items into [E] ones. Each row states the evidence that would promote it.

Q. A17. In-situ process monitoring for digital defect acceptance

The “digital qualification” direction of Section VII becomes concrete when in-situ monitoring is tied to μ CT-validated detection performance and then to the corrosion defect criteria of Section IX-C. Evidence state of the art [E]: layerwise-imaging CNNs detect LOF against μ CT ground truth at >93% accuracy; self-organizing-map models achieve 61–94% sensitivity and 69–93% specificity for LOF porosity; multi-layer signal fusion (photodiode + imaging) reaches 91.3% accuracy / 91.5% F1 for localized porosity at 800 μ m cluster resolution; deep belief networks classify porosity mode at up to 99.6%; synchrotron-ground-truth photodiode analyses resolve pore classes with a physically meaningful 15 μ m boundary between benign gas pores and structurally detrimental keyhole pores; SWIR thermography predicts voxel porosity at AUC ~0.86; and fringe-projection topography indicators predict local density drops with XCT validation [88]–[90]. Near-surface pore formation—the corrosion-relevant population—has specifically been characterized by combined melt-pool monitoring and XCT [91]; acoustic-emission monitoring is the emerging counterpart for WAAM.

TABLE I

REGISTRY OF ALL [P] ITEMS IN THIS REVIEW: HYPOTHESES OFFERED FOR COMMUNITY DISCUSSION AND VALIDATION—*none claimed as pilot practice by the authors at submission*. EACH ROW NAMES THE EVIDENCE THAT WOULD PROMOTE THE ITEM TO [E] STATUS.

ID	Section	Proposal (all marked [P])	Promotion criterion (what would make it [E])
P1	A1	Minimum reporting standard for AM-HEA electrochemistry (Ag/AgCl/seawater, G1 surface prep, $n \geq 3$, co-tested references)	Adoption by ≥ 3 independent groups with cross-laboratory reproducibility
P2	A11	Interlaboratory benchmark protocol (3 labs $\times$ 2 powder lots $\times$ 3 builds $\times$ 2 orientations $\times$ 2 finishes)	Completed round-robin with bounded lab-to-lab variance
P3	A12	Galvanic acceptance: $\leq 10\%$ rate increase over uncoupled substrate; three-potential CP test (free, -0.80 , -1.10 V vs Ag/AgCl)	Measured HEA–steel and HEA–bronze couples incl. CP arms
P4	A3	Defect acceptance: near-surface LOF class, max-pore-size limits, inclusion density, small-electrode initiation statistics	Correlated defect→pit datasets on AM HEAs (currently analog-only)
P5	A5	Natural-seawater exposure matrix (zones $\times$ 1/6/12/24 mo $\times$ crevice assemblies, ≥ 2 sites)	Completed exposures with G46/G1 endpoint reports
P6	A7	EHEA post-processing sequence: HIP $\rightarrow$ machine $\rightarrow$ EP/laser-polish; avoid homogenization anneals	EHEA-specific finishing studies with passivity re-qualification
P7	A13	WAAM over-alloying bands (Mn +2–5, Cr +1–3 wt%) and drift acceptance ($\pm 10\%$ Mn rel.)	Measured wire→deposit balances across lots and processes
P8	A14/A18	Indicative cost model and TCO breakeven scaling (downtime $\gtrsim \$1$ – 2 k/event $\Rightarrow$ match incumbent interval)	Real quotations and field interval data
P9	A15	Dual-torque MCA protocol (0.28 and 2–3 N·m) with site-count statistics on downskin surfaces	Reproducible crevice statistics on AM HEAs
P10	A9	MIC first-line set and endpoints (pit-depth ratio ≤ 1.5 , R_p retention $\geq 70\%$, TM0194/TM0212 monitoring)	Completed AM-HEA MIC program (any organism set)
P11	A17	Monitoring POD targets (LOF ≥ 100 μm : POD ≥ 0.9 ; ≥ 25 μm in 200 μm shell: POD ≥ 0.9 ; FDR $\leq 5\%$)	μCT -validated detection studies on HEA chemistries
P12	A6	ICME pipeline: CALPHAD/CSI + passivity surrogates + Pareto screening, DED-library validation	Printed combinatorial coupons with measured passivity

Target detection metrics for corrosion-service acceptance [P]: (i) probability of detection ≥ 0.90 , validated against μCT , for LOF ≥ 100 μm anywhere in the build and for pores ≥ 25 μm within the 200 μm corrosion-critical near-surface shell (matching the Section IX-C classification); (ii) false-discovery rate $\leq 5\%$ at those thresholds; (iii) per-build μCT verification of at least one sentinel coupon including a near-surface scan, to re-anchor the monitoring model (chemistry changes shift melt-pool signatures, so HEA models must not be transferred from 316L data without retraining); (iv) voxel-registered defect maps folded into the build-data package per DNV-ST-B203 [42], with corrosion-critical zones gated more tightly than interior zones; (v) periodic blind-challenge audits of the classifier with deliberately off-window layers. Until such POD evidence exists, monitoring supplements but does not replace μCT -based acceptance.

R. A18. Life-cycle cost scenario: brackish-water impeller with fouling and erosion

Table II works the TCO logic of Section IX-N for a brackish-water pump impeller (lower salinity than open ocean, so general corrosion is mild while biofouling and silt erosion dominate maintenance). Assumptions [P]: 25-year horizon, zero discount (for clarity), bronze purchase \$2.5k with 6-year service interval (events at years 0/6/12/18/24), WAAM-HEA first cost \$3k (Table VIII), downtime-plus-labor D per replacement event, three illustrative D levels. Result: with downtime worth $\geq \$2$ k per event, an AM-HEA impeller that merely *matches* the 6-year bronze interval already wins on TCO (and wins by a wide margin at \$8k/event); it loses only if its interval falls *below* the incumbent’s when downtime is cheap. The scaling rule for substitution is therefore: *required HEA service interval $\approx$ incumbent interval for $D \gtrsim \$1$ – 2 k per event;*

≈ 1.2 – $1.3 \times$ incumbent interval when downtime is negligible. LCA reading [P]: every avoided replacement amortizes the HEA’s critical-element (Co) burden and the atomization energy; conversely, if field intervals do not at least match bronze, both TCO and LCA favor the incumbent. Fouling/erosion specifics: an electropolished HEA surface plausibly reduces cleaning frequency (biofouling-release analogy; unproven for HEAs [2]), and EHEA/FCC strength-cavitation resistance arguments support the erosion assumption—but both are evidence gates, not credits, until the Section IX-E exposure data exist.

X. TABLES AND FIGURES

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TABLE II

TOTAL-COST-OF-OWNERSHIP SCENARIO: BRACKISH-WATER PUMP IMPELLER, 25-YEAR HORIZON, ZERO DISCOUNT RATE. COSTS [P] EXCEPT ANCHORS IN TABLE VIII. BRONZE BASELINE: \$2.5K PURCHASE, 6-YR INTERVAL (5 PURCHASES; 4 DOWNTIME EVENTS). WAAM-HEA: \$3K FIRST COST, INTERVAL T . D = DOWNTIME-PLUS-LABOR PER REPLACEMENT EVENT.

Downtime D per event	Bronze TCO (25 yr)	HEA TCO at $T = 6$ yr	HEA TCO at $T = 12$ yr	Breakeven rule for HEA
\$8k (critical service)	$12.5 + 4 \times 8 = \$44.5k$	$3 + 3 \times 11 = \$36k$	$3 + 1 \times 11 = \$14k$	wins if $T \geq \sim 5$ yr (below bronze interval)
\$2k (typical berth)	$12.5 + 4 \times 2 = \$20.5k$	$3 + 3 \times 5 = \$18k$	$3 + 1 \times 5 = \$8k$	wins if $T \geq 6$ yr (match bronze)
\$0.5k (non-critical)	$12.5 + 4 \times 0.5 = \$14.5k$	$3 + 3 \times 3.5 = \$13.5k$	$3 + 1 \times 3.5 = \$6.5k$	wins if $T \geq \sim 6$ yr; margin thin

Scaling rule [P]: required HEA interval $\approx$ incumbent interval when downtime $\gtrsim$ \$1–2k/event; ≈ 1.2 – $1.3\times$ incumbent when downtime is negligible. LCA: each avoided replacement amortizes the Co-bearing feedstock burden and atomization energy; fouling-release benefit of electropolished HEA surfaces is plausible but unproven [2].

TABLE III

QUANTITATIVE ELECTROCHEMICAL SYNTHESIS FOR THE MAIN AM-HEA FAMILIES IN CHLORIDE MEDIA, WITH 316L BENCHMARKS. POTENTIALS ARE QUOTED AS PUBLISHED (REFERENCE ELECTRODES DIFFER BETWEEN STUDIES: SCE OR Ag/AgCl; ABSOLUTE VALUES SHIFT BY TENS OF mV UNDER CONVERSION). “AB” = AS-BUILT. RANGES ACROSS STUDIES ARE THE HONEST SUMMARY STATISTIC; SEE SECTION IX-A.

System	Route / state	Medium	E_{corr}	i_{corr} ($\mu\text{A}/\text{cm}^2$)	E_{pit} / key metric
CoCrFeMnNi	SLM, AB	3.5% NaCl	−189 mV	0.09	+197 mV; passivity
CoCrFeMnNi (cast ref.)	Cast	3.5% NaCl	−179 mV	0.11	+37 mV; window 2
CoCrFeMnNi	SLM, XY / XZ	3.5% NaCl	−0.346 / −0.323 V vs SCE	0.123 / 0.133	Cast best: −0.264 V
CoCrFeMnNi	LPBF, AB + HIP	3.5% NaCl	—	—	$\sim 2.2\times$ better than 3
CoCrFeMnNi (friction-surfaced)	Solid-state deposit	3.5% NaCl	—	0.11 (cast: 0.55)	+125.7 mV (cast: −
CoCrNi	LAAM	3.5% NaCl	—	lower with more oxides	No pitting in test wi
AlCoCrFeNi _{2.1}	SLM, AB	3.5% NaCl	—	—	Best of as-built/anne
AlCoCrFeNi _{2.1}	Cast	3.5% NaCl (+NaHSO ₃)	—	—	Passivates 5–45 °C;
AlCoCrFeNi _{2.1} Mo _{0.3}	Cast (AM design target)	3.5% NaCl	—	low passive current	E_{pit} + 282 mV; Mo
316L (AM benchmark)	SLM	1 M NaCl	—	—	CPT 27.5–31 °C vs

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TABLE IV
DEFECT CLASSES AND PIT-INITIATION EVIDENCE (AM STAINLESS STEELS; TRANSFER TO HEAS BY ANALOGY). BASIS FOR PROPOSED DEFECT ACCEPTANCE CRITERIA IN SECTION IX-C.

Defect class	Quantitative link to localized corrosion	Consequence for acceptance criteria
Lack-of-fusion (LOF) pores	LPBF 420 SS: 18 μm avg LOF at 0.42% porosity $\Rightarrow$ ~ 100 μm pits at defect zones vs < 2 μm defect-free ($\sim 50\times$ amplification) [52]; LOF are oxide-lined, spatter-derived pre-formed crevices [54]; minimizing LOF in DED 304L significantly improves corrosion response [53]	Class surface-near (< 200 μm) LOF separately from interior porosity; maximum-pore-size limits, not volume fraction alone
Gas porosity (total volume)	SLM 316L: 0.01–0.4% porosity does not shift E_{pit} but raises metastable-pitting frequency and lowers repassivation potential [53]	Control porosity for initiation-statistics and crevice-like behavior, not just the thermodynamic pitting threshold
Inclusions (MnS, oxides)	MnS and LOF jointly identified as initiation sites in sintered 316L; site density quantified via electrode-area scaling [56]; MnS/oxides present in WAAM Cantor deposits [33]; refined/absent MnS is why LPBF 316L can beat wrought [55]	Inclusion density limits for WAAM; exploit inclusion refinement in powder-bed routes
Surface roughness / as-built skin	10–30 μm Sa typical as-built LPBF 316L; lower roughness $\Rightarrow$ higher breakdown potential; top surfaces best, downskin worst [55]; rough as-built surfaces raise SCC susceptibility [3]	Specify the wetted-surface condition (machined, electropolished or laser-polished) as part of the material state
Build-plate location	Specimens downstream of shielding-gas flow show least corrosion resistance in LPBF 304L [53]	Map properties across the build plate during qualification, not only at coupon locations

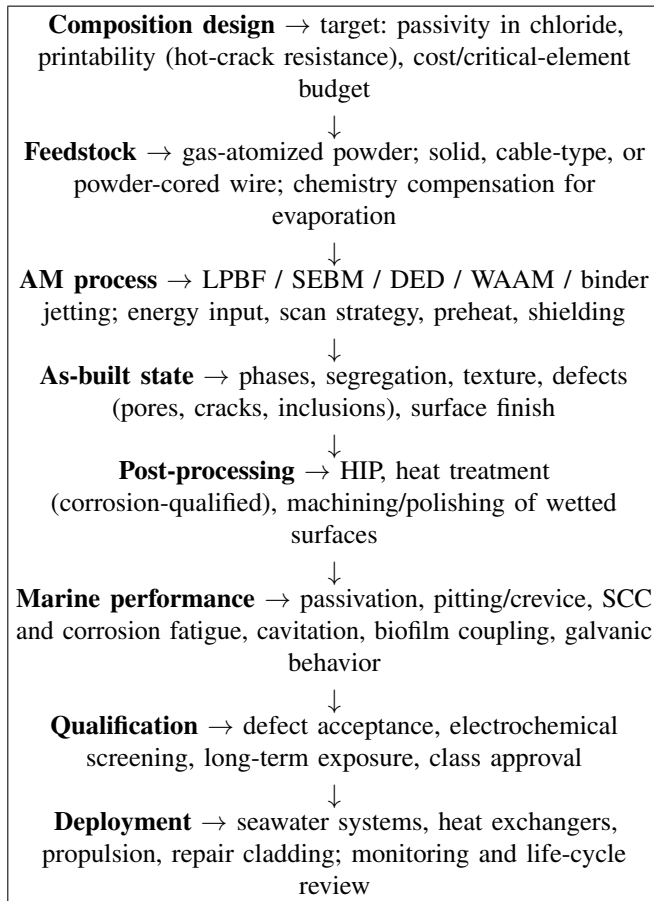


Fig. 1. Process–structure–property–qualification chain for AM HEAs in marine service. The framework organizes the review: each stage is a control point, and the weakest link (long-term seawater evidence and standards coverage) defines the deployment roadmap.

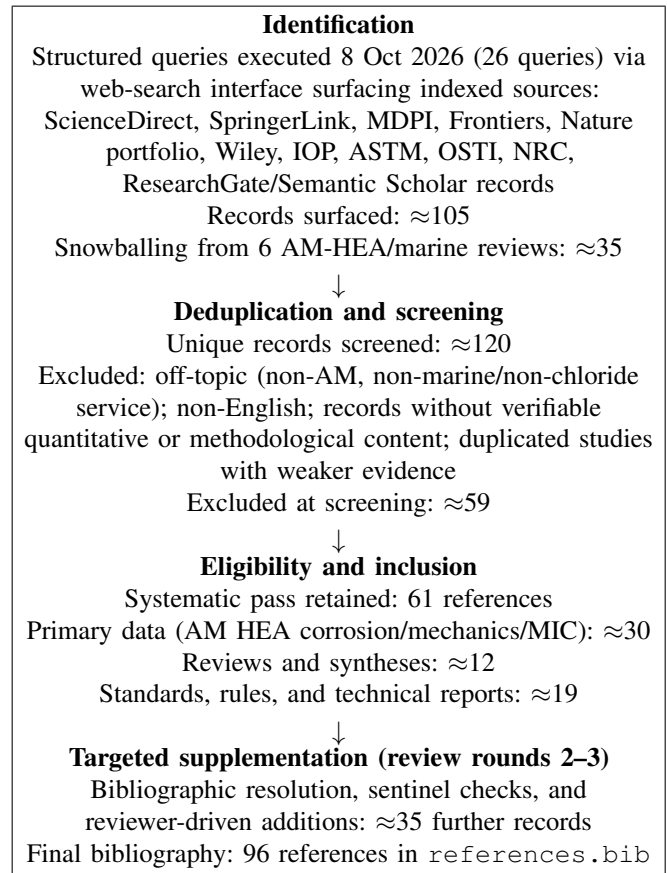


Fig. 2. PRISMA-style flow for this review (counts as recorded during the search; approximate). Limitations declared in Section I-A: no subscription-database access (Scopus/Web of Science), so coverage is structured but not exhaustive, and gray-literature records were included only when bibliographically verifiable.

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TABLE V

MINIMAL NATURAL-SEAWATER EXPOSURE MATRIX FOR INITIAL QUALIFICATION OF AM HEAS, STRUCTURED ON ASTM G52 [59], WITH CREVICE ASSEMBLIES PER ASTM G78 [60]. ALL CELLS: TRIPPLICATE SPECIMENS; TWO SURFACE CONDITIONS (AS-BUILT, MACHINED/POLISHED); TWO BUILD ORIENTATIONS; REFERENCE ALLOYS (LPBF 316L, WROUGHT 316L, NIAL BRONZE, 2205) IN IDENTICAL RACKS; AT LEAST TWO SITES WITH DIFFERENT TEMPERATURE/SALINITY/BIOTA.

Zone	Configuration	Durations	Flow condition	Primary endpoints
Atmospheric / splash	Flat panels, 90° racks [59]	1, 6, 12, 24 mo	Wet-dry cycling, natural splash	Mass loss (G1/G31 [49], [61]), pit depth (G46 [62]), rust/film analysis
Tidal	Panels on tidal rack [59]	1, 6, 12, 24 mo	Semidiurnal immersion cycles	Pit statistics, biofilm sampling, retained tensile properties
Full immersion (1–3 m)	Panels + crevice assemblies (G78 washers) + HEA-steel couples [57], [60]	1, 6, 12, 24 mo	Quiescent to local tidal flow	Crevice attack rating, galvanic current log, EIS at retrieval intervals
Velocity loop (laboratory)	Pumped seawater rig	3–6 mo	0.5–3 m/s (pump/heat-exchanger service)	Erosion-corrosion, passive-film stability under flow

High-velocity service is outside G52's scope and must be added as a separate loop. Endpoints at every retrieval: mass loss, maximum pit depth and rating, crevice inspection, retained mechanical properties, and biofilm/under-film sampling (Table VII).

TABLE VI

POST-PROCESSING BALANCE FOR EHEA AND FCC-HEA MARINE PARTS: DEFECT MITIGATION VERSUS SEGREGATION/PASSIVITY TRADE-OFFS.

Treatment	Effect on defects/segregation	Measured effect on corrosion	Recommendation for marine EHEA/FCC parts
HIP	Closes internal porosity; near-isothermal, does not deliberately drive segregation	HIPed LPBF CoCrFeMnNi keeps AB-level chloride performance; hardness 229→152 HV [30]	Use when porosity is the concern; safest densification
Homogenization anneal	Relieves stress but restores FCC/B2 segregation in EHEAs	SLM AlCoCrFeNi _{2.1} : as-built best; annealing dramatically degrades chloride passivity [11]	Avoid for passivity-critical parts; if required, re-qualify electrochemically
CW laser polishing	Localized remelting; Sa 2.1→0.3 μm (−85%), grain refinement −34%	i_{corr} −97%; passive window +98% (LPBF 316L) [66]; NS polishing risks microcracks, FS limited benefit [67]	Preferred for accessible wetted faces of LPBF parts
Electropolishing	Removes surface peaks/reaction layer; reaches internal channels	Cr/Fe of passive film $\sim 3\times$; i_{corr} 2.5→0.8 $\mu\text{A}/\text{cm}^2$ (316L) [68]	Preferred for conformal/internal seawater channels

Recommended sequence: HIP → machining → electropolish or CW laser polish of wetted surfaces → low-temperature stress relief only if needed (with corrosion re-qualification).

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TABLE VII

STATUS OF HEA BIOFILM/MIC EVIDENCE AND A PROPOSED PROGRAM FOR AM HEAS. EXISTING STUDIES ARE ON CAST/WROUGHT MATERIAL IN SIMULATED MEDIA; NO AM-HEA NATURAL-SEAWATER BIOFILM STUDY EXISTS YET.

Element	Design
Existing evidence	<i>P. aeruginosa</i> on FeCoCrNiMo _{0.1} : passive film thinned, selective Fe dissolution via EET (first HEA MIC study) [18]; on Cantor-type HEA: max pit 4.77 vs 3.16 μm (14 d), OCP $-367 \rightarrow -500$ mV [19]; <i>D. vulgaris</i> (SRB) on CoCrFeMnNi: sulfides, film thinning, more aggressive under carbon starvation [20]
Materials	LPBF CoCrFeMnNi, SLM AlCoCrFeNi _{2.1} , and one WAAM deposit; as-built vs machined; LPBF 316L and 2205 as controls
Organisms/media	<i>P. aeruginosa</i> (aerobic model), <i>D. vulgaris</i> (SRB), natural-seawater biofilm controls; 2216E and ASTM D1141 media [35]; carbon-starved variants
Electrochemistry	OCP, LPR and EIS time series over 14–28 days, inoculated vs sterile (per [19], [20] protocols); $n \geq 3$
Surface/film analysis	CLSM live/dead imaging, SEM pit mapping, XPS/AES depth profiles of passive film, SKPFM micro-galvanic maps
Under-film chemistry	pH and O ₂ microelectrodes beneath biofilms at defined sites; sulfide assays for SRB cells
Integration	Couple one coupon set into the natural-seawater racks of Table V so laboratory MIC and field biofilm data converge
Acceptance endpoints (A9)	Max pit-depth ratio (biotic/sterile) ≤ 1.5 (anchored to measured $1.51 \times$ [19]); R_p retention $\geq 70\%$ at 14 d; no E_b below sterile control; G46 pit density [62]; sessile counts per NACE TM0194 [72]; plus an inoculated arm at imposed CP potential [74]

TABLE VIII

INDICATIVE FIRST-COST MODEL FOR A REPRESENTATIVE SEAWATER PUMP IMPELLER (~ 8 KG NET MASS) IN 2026 USD. ENTRIES MARKED [E] ARE ANCHORED TO PUBLISHED MARKET DATA; ENTRIES MARKED [P] ARE ESTIMATES/ASSUMPTIONS OF THIS WORK. MACHINE-HOUR RATES AND HEA FEEDSTOCK PRICES DOMINATE ALL SCENARIOS; FIGURES ARE ORDER-OF-MAGNITUDE, NOT QUOTATIONS.

Option	Feedstock [E/P]	Process time and rate [E/P]	Post-processing [P]	Indicative first cost
Incumbent: NiAl bronze casting	Casting $\sim \$25$ – $40/\text{kg}$ [P]	Pattern/mold + pour [P]	Trim, machine, balance	$\sim \$2$ – 3k ; replacement/refurbish every ~ 5 – 8 yr in abrasive-fouled service [P]
LPBF HEA monolith	HEA powder $\sim \$150$ – $400/\text{kg}$ [P]; 316L powder anchor $\$20$ – $40/\text{kg}$ [E] [81], [82]; ~ 6 kg effective at 90% reuse	~ 30 h at $\$50$ – $150/\text{h}$ [E] [82], [83]	HIP, machine, electropolish	$\sim \$6$ – 7k
WAAM HEA + machining	HEA MPCW wire $\sim \$40$ – $120/\text{kg}$ [P]; ~ 12 kg	~ 10 h at $\$15$ – $40/\text{h}$ [P]	Machining ($\sim 40\%$ mass removal), surface finish	$\sim \$2.5$ – 3k
DED HEA cladding on 316L/steel	Substrate casting $\sim \$800$ [P]; overlay powder ~ 11 kg at $\$150$ – $250/\text{kg}$ [P]	~ 8 h at $\$30$ – $80/\text{h}$ [P]	Machine to profile, local polish	$\sim \$4$ – 5k

Life-cycle notes [P]: AM buy-to-fly advantages (near-net deposition vs machining from bar/forging), extended maintenance intervals if HEA pit/cavitation resistance holds in field service (unproven—evidence gate in Fig. 3), and critical-element burden (10–20% Co in FCC HEAs) must be weighed in the LCA against Cu-based incumbents [2].

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TABLE IX
ADDITIVE MANUFACTURING ROUTES FOR HEAS AND THEIR MARINE RELEVANCE (SYNTHESIZED FROM [12], [13], [15], [16]).

Process	Solidification and microstructure	Principal advantages	Principal limitations	Marine relevance
LPBF (SLM)	Very high cooling rate ($\sim 10^5$ – 10^6 K/s); fine cells, high dislocation density, build-direction texture	High resolution, complex geometry, best as-built homogeneity; fine parts	Small build volume, slow rate, porosity/cracking off-window, powder cost	Impellers, valves, fittings, heat-exchanger cores; complex seawater-system parts
SEBM	High cooling rate under vacuum; bed preheat; slightly coarser than LPBF	Low residual stress, good for crack-prone and refractory HEAs	Coarser surface, equipment cost, limited material library	Crack-sensitive or refractory compositions; parts later machined to wetted finish
DED / LMD	Moderate-to-high cooling; variable microstructure along build; in-situ alloying	Repair and cladding, graded deposits, high deposition, build on existing parts	Lower accuracy, dilution control, surface finish	Overlays on shafts, seal seats, rudder stocks; steel-to-HEA graded construction
WAAM	Low-to-moderate cooling; coarse columnar grains; interpass reheating	Meter-scale builds, high rate, low equipment cost, shipyard-compatible	Wire feedstock immaturity, Mn evaporation, inclusions, anisotropy	Propeller blanks, brackets, off-shore nodes, large repairs (machined to finish)
Binder jetting	No melting; sintered/infiltrated microstructure	Fast, cheap, batch production	Residual porosity unless infiltrated/HIPed; lower strength	Secondary route: porous/functional parts; limited for wetted load-bearing service

TABLE X
SELECTED CORROSION EVIDENCE FOR AM HEAS IN CHLORIDE MEDIA. ALL ENTRIES ARE LABORATORY SALINE TESTS UNLESS NOTED; LONG-TERM NATURAL SEAWATER DATA FOR AM HEAS REMAIN UNAVAILABLE.

Alloy	AM route	Medium	Key finding
CoCrFeMnNi	LPBF	3.5% NaCl	Higher pitting resistance and polarization resistance than cast; $\sim 58\%$ lower
CoCrFeMnNi	LPBF (as-built, HIPed)	3.5% NaCl vs. H_2SO_4	$\sim 2.2\times$ better than 316L in chloride; $\sim 15.5\times$ worse in acid; low H permea
AlCoCrFeNi _{2.1}	SLM	3.5% NaCl	As-built homogeneous duplex outperforms annealed and cast; annealing res
AlCoCrFeNi _{2.1} Mo _x	(cast; AM-adjacent design)	3.5% NaCl	Mo ($x=0.3$) raises strength and pitting potential via Mo/Cr composite oxid
AlMo _{0.25} FeCoCrNi _{2.1}	LDDED	Saline	Annealing heat treatment jointly tunes strength and corrosion response
AlCoCrFeNi (Ti, Cu doped)	LMD	Saline	Doping improves corrosion resistance over undoped; intermetallics must be
Nb–Mo–Ta–W	LPBF	Saline	Better overall than 316L; segregation sites produce pits up to ~ 1 mm deep
CoCrW	LPBF	Acidic NaCl, 28 d immersion	Oxide film matures then destabilizes; attack at melt-pool boundaries

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TABLE XI
CANDIDATE MARINE APPLICATIONS FOR AM HEAS, PREFERRED ROUTES, AND ENABLING RATIONALE.

Marine subsystem	Candidate components	Preferred AM route	Enabling rationale	Readiness
Seawater systems	Impellers, valve trim, strainers, fittings	LPBF	Complex geometry; chloride passivity of FCC/EHEA compositions [13], [30]	Lab/pilot
Heat exchangers, desalination	Conformal-channel exchangers, RO hardware	LPBF	Internal-channel freedom; pitting/crevice resistance governs life [13]	Lab/pilot
Propulsion	Propellers, shafting features	WAAM (+ machining)	Meter-scale deposition; shipyard-compatible [15]	Lab/demonstrator
Structure and outfitting	Brackets, nodal repairs	WAAM / DED	Large builds; cost-effective deposition [15]	Lab/demonstrator
Repair and maintenance	Shaft, seal-seat, rudder-stock overlays	DED / LMD	Localized corrosion/wear resistance; graded interfaces [14]	Pilot (steel substrates)
Hot/abrasive niches	Exhaust-side hardware	SEBM, WAAM (RHEA)	High-temperature strength of refractory HEAs [16], [31]	Lab

TABLE XII
EXPANDED STANDARDS AND RULES FRAMEWORK FOR QUALIFYING AM HEA COMPONENTS FOR MARINE SERVICE, COVERING ELECTROCHEMICAL SCREENING, PITTING/CREVICE, STRESS CORROSION CRACKING, GALVANIC COUPLING, AND FIELD EXPOSURE (SECTION IX-J).

Document	Scope	Role in AM-HEA marine qualification
ASTM D1141 [35] ASTM G1, G31 [49], [61] ASTM G3, G5, G61 [36]–[38]	Substitute ocean water preparation Specimen preparation/cleaning; immersion testing Electrochemical conventions; anodic and cyclic polarization	Standard medium for reproducible seawater tests Baseline mass-loss program and retrieval evaluation Screening of passivity, pitting and repassivation
ASTM G48 Methods A–F [86]	Ferric chloride pitting (A) and crevice (B) tests; critical pitting/crevice temperature for Ni alloys (C/D) and stainless steels (E/F)	Accelerated ranking; CPT/CCT as single-number acceptance metrics
ASTM G78 [60]	Crevice corrosion testing of stainless steels in marine applications (washer-type assemblies)	Crevice assemblies for laboratory and seawater racks
ASTM G192 [87]	Crevice repassivation potential (potentiodynamic–galvanostatic–potentiostatic)	Most discriminating single metric for crevice susceptibility
ASTM G46 [62] ASTM G30, G38/G39 [92]–[94] ASTM G129 (and G123) [95], [96]	Examination and rating of pitting U-bend and C-ring stress-corrosion specimens Slow strain rate testing for environmentally assisted cracking (boiling acidified NaCl for stainless alloys)	Pit-density and pit-depth statistics after exposure SCC screening under residual/applied stress in seawater Quantitative SCC susceptibility under load in chloride media
ASTM G71 [57]	Galvanic corrosion testing in electrolytes	HEA–steel and HEA–bronze couples; CP compatibility screening
ASTM G52 [59]	Exposure of metals in surface seawater (immersion, tidal, splash, spray; crevices per G78; controls per D3623)	Governing practice for the natural-exposure matrix (Table V)
ASTM G16 [75] DNV-ST-B203 [42]	Statistics applied to corrosion data Additive manufacturing of metallic parts (marine)	Required statistical treatment of pit and exposure data Qualification framework to extend toward HEA compositions and defect acceptance
DNV-RP-B401 [58]	Cathodic protection design	CP set points against which HEA compatibility is judged (Section IX-D)
ABS metal AM guidelines [43] Lloyd’s Register AM requirements [44]	Class approval of AM metal parts Class approval of AM processes and parts	Parallel route for machinery and hull components Parallel route; data-driven qualification expectations

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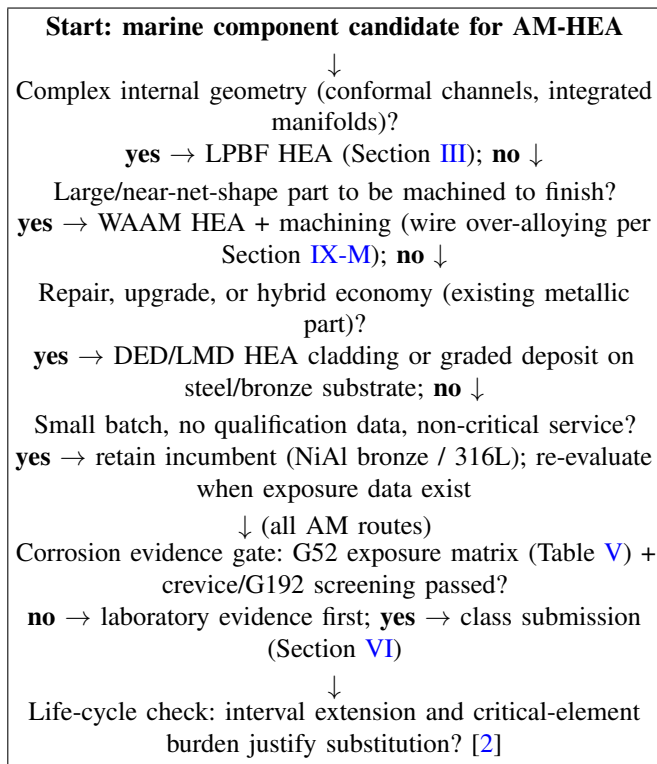


Fig. 3. Decision tree for route selection and deployment gating of AM-HEA marine components, linked to the indicative cost model (Table VIII). All decision logic is proposal [P]; the cost anchors behind it are flagged in Section IX-N.

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